

Conditional Fiber Geometry for Incomplete Inverse Data

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Abstract

Let $V \in E \subset \mathbb{R}^d$ parameterize a finite-dimensional inverse problem, let $X = F(V)$ denote its complete data, and let a mask M reveal $P_M X$. We study the conditional geometry induced by this incomplete observation. Near a compact regular observation fiber with Jacobian rank r , the ambient level set has dimension $d - r$, and the support of the associated conditional data law has covering upper exponent at most $d - r$. If the restricted differential $DF|_{\ker DH}$ has constant rank ℓ on a relative neighborhood of the fiber in the regular level set, the image exponent can be sharpened to ℓ ; because $H = PF$, this equals $\text{rank } DF - \text{rank } DH$ when both ambient ranks are constant. For noisy observations, we derive local covering estimates that retain every nonzero singular value of the masked, noise-whitened Jacobian. On a compact regular region satisfying boundary transversality and a bidirectional no-exit condition, a pseudoinverse horizontal flow transports complete fibers with displacement bounded by $\|z - z'\|/s_0$. Coarea disintegration then yields quantitative total-variation control after transport and Wasserstein-1 continuity of the conditional data laws. We further establish motion and curvature estimates for noncollapsed image manifolds and L^1 regularity after Gaussian smoothing. Experiments on a discretized quadrilateral EIT model examine these mechanisms independently. Two fiber-construction procedures recover PCA dimensions 6, 4, and 2 for numerical ranks 2, 4, and 6; a separate rank-eight experiment returns to the same numerical solution, consistent with local isolation. Additional tests support the predicted δ/σ_i scaling and the use of the full singular spectrum. In a separate three-seed held-out study, a mask selected by a training-only spectral score reduced conditional EDM completion error relative to 20 score-trimmed random equal-budget controls. This is engineering evidence, not a test of the external DDPM rate. The numerical results do not establish differential rank, global flow completeness for the polygon class, continuum finite-element convergence, conditional-score regularity, or a uniform diffusion-model rate.

1 Introduction

Incomplete measurements constrain an inverse-problem parameter without, in general, identifying it. The remaining feasible parameters form an observation fiber, and its image under the complete-data map contains the support of the corresponding conditional data law. This elementary description becomes useful when the complexity of a conditional generative model is controlled through metric entropy. It also separates two properties that measurement count alone cannot distinguish: residual tangent dimension and stability in the observed normal directions.

This work is motivated by polygonal electrical impedance tomography (EIT), where data completion has been studied both as a computational device [5] and through conditional diffusion models [6]. Finite-measurement identifiability and stability depend on the admissible conductivity class and measurement configuration [3, 7]. Jacobian singular values, Fisher information, sensitivity-volume criteria, and Bayesian experimental design are established tools in EIT [8, 9, 12, 13]. Conditional distributions on regular level sets are likewise classical consequences of disintegration

and the coarea formula and have recently been used explicitly in conditional generative modeling [1]. These ingredients are used here as background, not presented as new constructions.

Connections among local identifiability, active subspaces, and parameter-space reduction have also been studied explicitly [4]. Separate minimax work analyzes manifold adaptation for conditional diffusion estimation [16]; its hypotheses and rates are not used in the theorem below.

The EIT analysis of Chen et al. [6] controls the conditional law by the entropy of the global polygonal Dirichlet-to-Neumann (DtN) class. Its conditional diffusion bound therefore retains dimension $2n_v$ for every observation of an n_v -vertex polygon. We replace that global set by

$$F(E \cap (P_m F)^{-1}(z)),$$

the image of the actual observation fiber. The resulting analysis connects masked inverse observations, fiber entropy, and the intrinsic covering input used by an external diffusion convergence theorem.

The contributions are as follows.

1. We derive compact regular-fiber covering bounds, sharpen them using the rank of DF restricted to the fiber tangent bundle, and formulate the corresponding conditional-support result for a random family of masks.
2. We establish linear and local nonlinear noisy-tube bounds that retain the complete nonzero singular spectrum of the observation Jacobian, including its noise-whitened form.
3. Under explicit boundary and flow-completeness assumptions, we construct a quantitative pseudoinverse transport between fibers and prove stability of the coarea conditional measures and their complete-data pushforwards.
4. We apply the entropy bound pointwise to the version-pinned theorem of Liang et al. [11, arXiv version 2]. This is a conditional corollary of an external result, not an observation-uniform convergence theorem.
5. We provide numerical EIT experiments designed to test the local geometric mechanisms and their failure modes in a finite-dimensional discretization.

All theoretical laws below are population laws. A finite empirical training set has finite support and hence zero covering exponent below its minimum sample separation; it is not the object of the continuum entropy theorem. The abstract results hold under the stated hypotheses. Their global validity over the complete polygon prior used in EIT is not asserted.

Section 2 introduces the mathematical setting. Sections 3 and 4 treat exact fibers and noisy tubes. Sections 5 and 6 develop fiber and measure transport, Section 7 considers image manifolds and Gaussian smoothing, and Section 8 states the external diffusion consequence. Sections 9 and 10 describe the EIT model, numerical evidence, and supporting interface checks.

2 Mathematical setting

Let $O \subset \mathbb{R}^d$ be open, $E \Subset O$ compact, and $F \in C^1(O; \mathbb{R}^D)$. A finite or countable mask family $\mathcal{M}$ contains linear maps $P_m : \mathbb{R}^D \rightarrow \mathbb{R}^{q_m}$. Put

$$H_m = P_m F, \quad S_{m,z} = E \cap H_m^{-1}(z).$$

Equip the countable disjoint union

$$Z = \bigsqcup_{m \in \mathcal{M}} (\{m\} \times \mathbb{R}^{q_m})$$

with its standard Borel structure. Let (V, M) be any jointly distributed $E \times \mathcal{M}$ -valued random element; independence is unnecessary. Define

$$X = F(V), \quad Y = P_M X, \quad Z = (M, Y) \in \mathcal{Z}.$$

Covering numbers $\mathcal{N}_\varepsilon(A)$ use Euclidean balls whose centers may lie in the ambient space; requiring centers in A changes only fixed radius constants. We use $\text{TV}(\mu, \nu) = \sup_B |\mu(B) - \nu(B)|$, equivalently one half of the L^1 distance when densities exist. All singular values refer to explicitly fixed Euclidean parameter and observation norms. Under noise covariance Γ , the observation norm is whitened and the relevant Jacobian is $\Gamma^{-1/2} DH$.

3 Conditional support and exact-fiber entropy

Theorem 3.1 (Compact regular-fiber entropy). *Fix a linear $P : \mathbb{R}^D \rightarrow \mathbb{R}^q$, put $H = PF$, and let $S_z = E \cap H^{-1}(z)$ be nonempty. Suppose an open U satisfies $S_z \subset U \subset O$ and $\text{rank } DH(v) = r$ for every $v \in U$. Set $k = d - r$. Then $H^{-1}(z) \cap U$ is an embedded C^1 submanifold of dimension k . If $k \geq 1$, there are $C_z < \infty$ and $\varepsilon_z > 0$ such that*

$$\mathcal{N}_\varepsilon(F(S_z)) \leq C_z \varepsilon^{-k}, \quad 0 < \varepsilon \leq \varepsilon_z. \quad (1)$$

If $k = 0$, S_z and $F(S_z)$ are finite.

Proof. The constant-rank level-set theorem makes $Q_z = H^{-1}(z) \cap U$ an embedded C^1 manifold of dimension k [10]. Every $v \in S_z$ has a relatively compact chart whose inverse parametrization $\psi_v : B_k(0, a_v) \rightarrow Q_z$ has bounded derivative on a closed smaller ball. The composition $F \circ \psi_v$ is C^1 there; its derivative is bounded on that closed Euclidean ball, which is convex, so the composition is Lipschitz. Compactness of S_z selects finitely many charts, indexed by $j = 1, \dots, L$. If K_j is the resulting Lipschitz constant, the volumetric Euclidean bound gives

$$\mathcal{N}_\varepsilon(F(S_z \cap \psi_j(B_k(0, a_j)))) \leq \left(1 + \frac{2a_j K_j}{\varepsilon}\right)^k.$$

Summing the finite covers proves (1) after restricting ε . For $k = 0$, Q_z is discrete, and its compact subset S_z is finite. $\square$

Corollary 3.2 (Image-rank-refined fiber entropy). *In the setting of Theorem 3.1, let $Q_z = H^{-1}(z) \cap U$ and suppose a relatively open neighborhood $W \subset Q_z$ of S_z satisfies*

$$\text{rank}(DF(v)|_{\ker DH(v)}) = \ell, \quad v \in W.$$

If $\ell \geq 1$, there are $C_{z,\text{im}} < \infty$ and $\varepsilon_{z,\text{im}} > 0$ such that

$$\mathcal{N}_\varepsilon(F(S_z)) \leq C_{z,\text{im}} \varepsilon^{-\ell}, \quad 0 < \varepsilon \leq \varepsilon_{z,\text{im}}. \quad (2)$$

If $\ell = 0$, $F(S_z)$ is finite, although S_z itself need not be finite. Moreover, since $H = PF$, if $\text{rank } DF(v) = s$ on W , the restricted-rank hypothesis holds automatically with

$$\ell = s - r. \quad (3)$$

Here the rank of DH is already r on W by Theorem 3.1.

Proof. The tangent space of Q_z is $T_v Q_z = \ker DH(v)$. Thus the restriction $F|_W$ has constant differential rank ℓ . The constant-rank theorem for maps between manifolds gives, around each point of S_z , a local C^1 factorization through $\mathbb{R}^\ell$. After shrinking to relatively compact coordinate neighborhoods, the factors have bounded derivatives on compact convex coordinate balls and are Lipschitz. Compactness of S_z selects finitely many such neighborhoods, and the same volumetric covering argument as in Theorem 3.1 gives (2). When $\ell = 0$, $F|_W$ is locally constant, so a finite subcover of S_z shows that $F(S_z)$ is finite.

Finally $DH = P DF$ implies $\ker DF(v) \subseteq \ker DH(v)$. Hence the kernel of $DF(v)|_{\ker DH(v)}$ is exactly $\ker DF(v)$, and rank-nullity gives

$$\text{rank}(DF(v)|_{\ker DH(v)}) = (d - r) - (d - s) = s - r.$$

□

Proposition 3.3 (Random-mask conditional support). *Let $\nu_{m,z}$ be a regular conditional distribution of V given $Z = (m, z)$. For $\mathcal{L}(Z)$ -almost every (m, z) ,*

$$\nu_{m,z}(S_{m,z}) = 1, \quad \text{supp } \mathcal{L}(X | M = m, Y = z) \subseteq F(S_{m,z}).$$

Proof. The parameter, mask, and observation spaces are standard Borel, so let $\kappa_{m,z}$ be a regular conditional distribution of (V, M) given Z . For the measurable map $T(v, m) = (m, P_m F(v))$, the identity $Z = T(V, M)$ implies $T_\# \kappa_{m,z} = \delta_{(m,z)}$ for $\mathcal{L}(Z)$ -almost every (m, z) . This follows first on a countable generator of the disjoint-union Borel sigma-field and then by the monotone-class theorem. Consequently $\kappa_{m,z}$ is concentrated on $\{(v, m) : H_m(v) = z\}$. Its V -marginal is $\nu_{m,z}$, so $\nu_{m,z}(S_{m,z}) = 1$. The conditional law of X is $F_\# \nu_{m,z}$; since $S_{m,z}$ is compact, $F(S_{m,z})$ is closed, proving the support inclusion. □

Corollary 3.4 (Random-mask conditional entropy). *Suppose $d = 2n_v$. Let $Z_{\text{reg}} \subset Z$ be a Borel set with $\mathcal{L}(Z)(Z_{\text{reg}}) = 1$ such that, for every $(m, z) \in Z_{\text{reg}}$, the nonempty compact fiber $S_{m,z}$ has an open neighborhood $U_{m,z}$ on which DH_m has constant rank $r(m, z)$. For $\mathcal{L}(Z)$ -almost every $(m, z) \in Z_{\text{reg}}$ with $r(m, z) < 2n_v$, a constant $C''_{m,z} < \infty$ satisfies*

$$\log \mathcal{N}_\varepsilon(\text{supp } \mathcal{L}(X | M = m, Y = z)) \leq C''_{m,z} + [2n_v - r(m, z)] \log(1/\varepsilon), \quad 0 < \varepsilon \leq 1.$$

If, additionally, a relatively open neighborhood of $S_{m,z}$ in $H_m^{-1}(z) \cap U_{m,z}$ has constant restricted rank

$$\text{rank}(DF(v)|_{\ker DH_m(v)}) = \ell(m, z),$$

then the same inequality holds with $\ell(m, z)$ in place of $2n_v - r(m, z)$. This additional hypothesis holds automatically with $\ell(m, z) = s(m, z) - r(m, z)$ if DF has constant rank $s(m, z)$ there. If either $r(m, z) = 2n_v$ or the certified image rank $\ell(m, z) = 0$, the conditional support is finite.

Proof. Combine Proposition 3.3 and Theorem 3.1; under the additional restricted-rank hypothesis use Corollary 3.2. Let ε_* be the minimum of 1 and the local scale supplied by the applicable covering bound. That result gives the stated form below ε_* after taking logarithms. On $[\varepsilon_*, 1]$, monotonicity gives $\log \mathcal{N}_\varepsilon \leq \log \mathcal{N}_{\varepsilon_*} < \infty$; enlarge the additive constant to cover this compact scale interval. □

Both exponents are upper bounds for an arbitrary compact intersection. The refined exponent can still be non-sharp because the conditional probability may occupy only part of the geometric image. No injectivity, embedded-image, or matching entropy lower bound follows from constant restricted rank alone.

4 Quantitative local geometry of noisy fibers

Theorem 4.1 (Linear full-spectrum tube). *Let $J \in \mathbb{R}^{q \times d}$ have rank $r \geq 1$ and nonzero singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$. For*

$$\mathcal{T}_{z,\delta}(R) = \{x \in B_d(0, R) : \|Jx - z\|_2 \leq \delta\},$$

if the set is nonempty, a constant C_d depending only on d satisfies

$$\mathcal{N}_\varepsilon(\mathcal{T}_{z,\delta}(R)) \leq \left(1 + \frac{C_d R}{\varepsilon}\right)^{d-r} \prod_{i=1}^r \left(1 + \frac{C_d \delta}{\sigma_i \varepsilon}\right), \quad \varepsilon > 0. \quad (4)$$

Proof. Take $J = U_r \Sigma_r V_r^\top$ and complete V_r by V_0 . Choose $x_0 \in \mathcal{T}_{z,\delta}(R)$. Write $x = V_r a + V_0 b$ and $x_0 = V_r a_0 + V_0 b_0$. Then $\|b\| \leq R$ and the two residual bounds imply

$$\|\Sigma_r(a - a_0)\|_2 = \|J(x - x_0)\|_2 \leq 2\delta.$$

Thus the tube lies in the product of a $(d - r)$ -ball of radius R and an ellipsoid with semiaxes $2\delta/\sigma_i$. Cover the two factors at radius $\varepsilon/\sqrt{2}$ and take products. Standard ball and axis-ellipsoid packing bounds [17, Chapter 4] give (4), after changing C_d . If centers must lie in the tube, use a maximal separated subset of the tube and bound it by the same containing-product packing number. $\square$

Corollary 4.2 (Noise-whitened tube). *For symmetric positive-definite Γ , replace the residual by $\|\Gamma^{-1/2}(Jx - z)\| \leq \delta$. Then Theorem 4.1 holds with the singular values of $\Gamma^{-1/2}J$.*

Proof. Apply Theorem 4.1 to $\tilde{J} = \Gamma^{-1/2}J$ and $\tilde{z} = \Gamma^{-1/2}z$. $\square$

When $\delta > 0$ and $\varepsilon \ll \delta/\sigma_i$ for every $i = 1, \dots, r$, the normal logarithmic contribution contains

$$r \log(\delta/\varepsilon) - \sum_{i=1}^r \log \sigma_i + O_d(1).$$

For full row rank, $\sum_i \log \sigma_i = \frac{1}{2} \log \det(JJ^\top)$. This does not introduce D-optimality; it gives the classical EIT sensitivity-volume criterion [13] a conditional-covering interpretation.

Lemma 4.3 (Quantitative implicit graph in singular coordinates). *Let $k = d - r$, let G be C^2 on a neighborhood of $\overline{B}_d(0, 2R)$, and suppose*

$$G(0, 0) = 0, \quad D_a G(0, 0) = \Sigma = \text{diag}(\sigma_1, \dots, \sigma_r), \quad D_b G(0, 0) = 0,$$

where $\sigma_1 \geq \dots \geq \sigma_r = s > 0$. Suppose also that

$$\|DG(x) - DG(y)\|_{\text{op}} \leq L\|x - y\|, \quad x, y \in \overline{B}_d(0, 2R), \quad LR \leq s/16.$$

Then there is a unique continuous map $g : \overline{B}_k(0, R) \rightarrow B_r(0, R/8)$ satisfying $G(g(b), b) = 0$; its restriction to $B_k(0, R)$ is C^1 . It obeys

$$\|g(b)\| \leq \frac{L}{s}\|b\|^2, \quad \|Dg(b)\|_{\text{op}} \leq \frac{17}{239} < \frac{1}{8}. \quad (5)$$

If $\|(a, b)\| \leq R$, $\|G(a, b)\| \leq \delta$, and $c = a - g(b)$, then

$$\|c\| \leq \frac{8\delta}{7s}, \quad \|\Sigma c\| \leq \frac{8\delta}{7}. \quad (6)$$

Proof. For fixed $\|b\| \leq R$, define $\mathcal{T}_b(a) = a - \Sigma^{-1}G(a, b)$ on $\overline{B}_r(0, R/8)$. If $\|a\| \leq R/8$, then $\|(a, b)\| < 17R/16$, and hence

$$\|D_a \mathcal{T}_b(a)\|_{\text{op}} \leq \frac{L}{s} \|(a, b)\| \leq \frac{17}{256}.$$

Taylor expansion along $t \mapsto (0, tb)$, using $D_b G(0, 0) = 0$, gives

$$\|G(0, b)\| \leq \frac{L}{2} \|b\|^2, \quad \|\mathcal{T}_b(0)\| \leq R/32.$$

Consequently

$$\|\mathcal{T}_b(a)\| \leq R/32 + (17/256)(R/8) < R/8.$$

The contraction theorem gives a unique fixed point $g(b)$ in this ball, and

$$\|g(b)\| \leq \frac{1}{1 - 17/256} \frac{L\|b\|^2}{2s} \leq \frac{L}{s} \|b\|^2 \leq R/16.$$

The parameter-dependent contraction theorem makes g continuously differentiable. At $(g(b), b)$, both $\|D_a G - \Sigma\|$ and $\|D_b G\|$ are at most $17LR/16$. Implicit differentiation therefore yields

$$\|Dg(b)\|_{\text{op}} \leq \frac{17LR/16}{s - 17LR/16} \leq \frac{17}{239}.$$

For the final assertion, the graph point $(g(b), b)$ lies in $B_d(0, 17R/16)$. The segment from it to (a, b) remains in that convex ball. The fundamental theorem of calculus gives

$$G(a, b) - G(g(b), b) = (\Sigma + E)c, \quad E = \int_0^1 [D_a G(g(b) + tc, b) - \Sigma] dt, \quad \|E\| \leq s/8.$$

Since $G(g(b), b) = 0$, the smallest singular value bound gives $\|c\| \leq 8\delta/(7s)$. Substitution back gives

$$\|\Sigma c\| \leq \|(\Sigma + E)c\| + \|E\|\|c\| \leq \delta + \delta/7 = 8\delta/7,$$

which proves (6). $\square$

Theorem 4.4 (Quantitative local nonlinear spectral tube). *Let $R > 0$. Suppose H is C^2 and F is C^1 on a neighborhood of $\overline{B}(v_0, 2R)$, with*

$$\sup_{\overline{B}(v_0, 2R)} \|D^2 H\|_{\text{op}} \leq L_H, \quad \sup_{\overline{B}(v_0, 2R)} \|DF\|_{\text{op}} \leq L_F.$$

Suppose $J_0 = DH(v_0)$ has rank at least $r \geq 1$, with leading singular values $\sigma_1 \geq \dots \geq \sigma_r = s > 0$, and assume

$$L_H R \leq s/16, \quad 0 \leq \delta \leq sR.$$

Then, for

$$Q_{v_0, \delta} = \overline{B}(v_0, R) \cap \{v : \|H(v) - H(v_0)\| \leq \delta\},$$

a constant C_d depending only on d satisfies, for every $\varepsilon > 0$,

$$\mathcal{N}_\varepsilon(F(Q_{v_0, \delta})) \leq \left(1 + \frac{C_d L_F R}{\varepsilon}\right)^{d-r} \prod_{i=1}^r \left(1 + \frac{C_d L_F \delta}{\sigma_i \varepsilon}\right). \quad (7)$$

No constant-rank assumption is required for this upper bound. If $\text{rank } J_0 > r$, the estimate resolves only the selected leading r constrained directions; the remaining nonzero directions are absorbed into the coarse $(d - r)$ factor, so the bound is intentionally not dimension-sharp. When $r = \text{rank } J_0$, the product contains the complete nonzero spectrum.

Proof. Choose the leading r singular triplets of J_0 , with orthonormal matrices U_r, V_r and $\Sigma_r = \text{diag}(\sigma_1, \dots, \sigma_r)$, so that $J_0 V_r = U_r \Sigma_r$ and $J_0^\top U_r = V_r \Sigma_r$. This is the leading rank- r truncated SVD block, not necessarily the full thin SVD. Complete V_r orthogonally by V_0 and define

$$G(a, b) = U_r^\top [H(v_0 + V_r a + V_0 b) - H(v_0)].$$

Then $D_a G(0, 0) = \Sigma_r$, $D_b G(0, 0) = 0$, and the orthogonal projection and coordinate changes preserve the derivative Lipschitz bound L_H . Apply Lemma 4.3. For every point of $Q_{v_0, \delta}$, $b \in \overline{B}_{d-r}(0, R)$ and, with $c = a - g(b)$,

$$c \in E_\delta := \{c : \|\Sigma_r c\| \leq 8\delta/7\}.$$

Thus $Q_{v_0, \delta}$ is contained in the image of

$$\Theta(b, c) = v_0 + V_r(g(b) + c) + V_0 b, \quad (b, c) \in \overline{B}_{d-r}(0, R) \times E_\delta.$$

The assumption $\delta \leq sR$ and (5)–(6) imply that this entire parametrized containing set lies in $B(v_0, 2R)$. Moreover, $\text{Lip}(\Theta) \leq 2$. The ball is convex, so the displayed derivative bound makes $F \circ \Theta$ $2L_F$ -Lipschitz there.

Cover $B_{d-r}(0, R)$ and the axis ellipsoid E_δ separately in the product Euclidean metric. Their standard volumetric covering bounds are a tangent factor of order $(1 + C_d R/\eta)^{d-r}$ and normal factors $\prod_i (1 + C_d \delta/(\sigma_i \eta))$. Take $\eta = \varepsilon/(2L_F)$; if $L_F = 0$, the image is a singleton and the claim is immediate. Mapping the product cover through $F \circ \Theta$ proves (7). If covering centers are required to belong to $F(Q_{v_0, \delta})$, select a point from each intersecting ambient cover ball and absorb the resulting factor two into C_d . $\square$

Corollary 4.5 (Uniform exact-fiber cover on a regular region). *Fix $1 \leq r \leq \min\{d, q\}$. Let $K \subset B_d(0, D)$ be compact and let $\rho_0 > 0$. Let U_0 be an open neighborhood of the closed ρ_0 -neighborhood of K . Suppose $H \in C^2(U_0; \mathbb{R}^q)$ and $F \in C^1(U_0; \mathbb{R}^D)$ satisfy uniform bounds $\|D^2 H\| \leq L_H$ and $\|DF\| \leq L_F$ there, and $\sigma_r(DH(v)) \geq s > 0$ uniformly on K . Put*

$$R = \min\{\rho_0/2, s/(16L_H)\},$$

with $s/(16L_H) = +\infty$ when $L_H = 0$. Then

$$\mathcal{N}_\varepsilon(F(K \cap H^{-1}(z))) \leq \left(1 + \frac{C_d D}{R}\right)^d \left(1 + \frac{C_d L_F R}{\varepsilon}\right)^{d-r}$$

for every $z \in \mathbb{R}^q$ and $\varepsilon > 0$.

Proof. Choose a maximal R -separated subset of the compact fiber. Its cardinality is at most $(1 + C_d D/R)^d$ by a volumetric packing bound. The associated closed R -balls cover the fiber. Apply Theorem 4.4 with $\delta = 0$ at every center and sum the local covers. $\square$

5 Quantitative transport of fibers

Definition 5.1 (Bidirectional no-exit property). Let H have full row rank on a neighborhood of $\mathcal{K}_G = \mathcal{K} \cap H^{-1}(G)$. The pair $(\mathcal{K}, G)$ has the bidirectional no-exit property for the minimum-norm horizontal lift if, for every ordered pair $z, z' \in G$ and every $v_0 \in S_z$, the initial-value problem

$$\dot{v}(t) = DH(v(t))^\dagger(z' - z), \quad v(0) = v_0,$$

has a unique solution for every $t \in [0, 1]$ and that solution remains in $\mathcal{K}$. Because the definition quantifies over ordered pairs, it also applies with z and z' exchanged. Under Assumption 5.2, Lemma 5.3 below shows that these solutions remain in $\mathcal{K}_G$ and satisfy $H(v(t)) = (1 - t)z + tz'$.

Assumption 5.2 (Quantitative regular observation tube). *Let $\mathcal{U} \subset \mathbb{R}^d$ be open, $\mathcal{K} \Subset \mathcal{U}$ a compact d -dimensional C^2 submanifold with boundary, and let $G \subset \mathbb{R}^r$ be nonempty, compact, and convex. Put $\mathcal{K}_G = \mathcal{K} \cap H^{-1}(G)$ and fix one open neighborhood $\mathcal{U}_Q$ of $\mathcal{K}_G$ with $\overline{\mathcal{U}_Q} \Subset \mathcal{U}$. All suprema and infima below are taken over this same fixed set $\mathcal{U}_Q$. Assume:*

1. $H \in C^2(\mathcal{U}; \mathbb{R}^r)$ has full row rank on $\mathcal{U}_Q$, $s_0 = \inf_{\mathcal{U}_Q} \sigma_r(DH) > 0$, and $L_H = \sup_{\mathcal{U}_Q} \|D^2H\|_{\text{op}} < \infty$;
2. the boundary restriction is also a submersion:
$$DH(v)(T_v \partial \mathcal{K}) = \mathbb{R}^r, \quad v \in \partial \mathcal{K} \cap H^{-1}(G).$$
3. $F \in C^2(\mathcal{U}; \mathbb{R}^D)$, with $M_F = \sup_{\mathcal{U}_Q} \|DF\|_{\text{op}}$ and $M_{F,2} = \sup_{\mathcal{U}_Q} \|D^2F\|_{\text{op}}$ finite;
4. V has a Lebesgue density ρ on $\mathcal{K}$ whose restriction to the regular tube extends to a positive C^1 function on $\mathcal{U}_Q$, with $0 < \rho_{\min} = \inf_{\mathcal{U}_Q} \rho \leq \sup_{\mathcal{U}_Q} \rho = \rho_{\max} < \infty$ and $L_\rho = \sup_{\mathcal{U}_Q} \|\nabla \rho\| < \infty$;
5. every $S_z = \mathcal{K} \cap H^{-1}(z)$, $z \in G$, is nonempty and has positive $(d-r)$ -dimensional Hausdorff measure;
6. $(\mathcal{K}, G)$ has the bidirectional no-exit property of Definition 5.1.

Write $k = d - r$, $D_{\mathcal{K}} = \text{diam}(\mathcal{K})$, and $D_F = \text{diam } F(\mathcal{K}_G)$.

The boundary-submersion hypothesis and the regular-level-set theorem with boundary make every S_z a neat C^2 submanifold of $\mathcal{K}$, with $\partial S_z = S_z \cap \partial \mathcal{K}$. It excludes boundary tangencies that can add isolated points to an otherwise positive-dimensional fiber. The no-exit property is a separate, essential flow-completeness hypothesis. The stronger condition

$$\text{range } DH(v)^\dagger \subseteq T_v \partial \mathcal{K}, \quad v \in \partial \mathcal{K} \cap H^{-1}(G),$$

is sufficient both for boundary submersion and for invariance of the boundary under every horizontal field, but it is not imposed when no-exit is verified by another argument.

The C^2 -boundary hypothesis is substantive. A polygon admissibility region cut out by several simultaneous side-length, angle, convexity, and separation inequalities will generally be a manifold with corners or a stratified set, not automatically a C^2 -boundary manifold. The transport results below therefore apply only after verifying Assumption 5.2 on a smooth full-dimensional compact subdomain with C^2 boundary, together with no-exit there. Lower-dimensional strata would require a separately formulated intrinsic theorem. No global extension across polygon corners or changes of active constraint is claimed.

Lemma 5.3 (Regular-tube bootstrap and ambient flow). *Under Assumption 5.2, every trajectory supplied by Definition 5.1 and starting in S_z remains in $\mathcal{K}_G$ for all $t \in [0, 1]$. For fixed $\xi = z' - z$, the vector field $b_\xi(v) = DH(v)^\dagger \xi$ is C^1 on $\mathcal{U}_Q$, and along each such trajectory its time-one solution is the restriction of an ambient C^1 local flow defined on a neighborhood of the initial point.*

Proof. Define

$$\tau = \sup\{t \in [0, 1] : v([0, t]) \subset \mathcal{U}_Q\}.$$

This number is positive because $v(0) \in S_z \subset \mathcal{K}_G \subset \mathcal{U}_Q$. On $[0, \tau)$, full row rank gives

$$\frac{d}{dt} H(v(t)) = DH(v(t)) DH(v(t))^\dagger \xi = \xi, \quad H(v(t)) = z + t\xi \in G.$$

If $\tau < 1$, continuity extends the displayed identity to $t = \tau$, while the state constraint in Definition 5.1 gives $v(\tau) \in \mathcal{K}$. Hence $H(v(\tau)) = z + \tau\xi \in G$ and $v(\tau) \in \mathcal{K}_G \subset \mathcal{U}_Q$. Openness of $\mathcal{U}_Q$ and continuity of v then extend the defining property of τ to a larger time, a contradiction. Therefore $\tau = 1$, the identity extends to the endpoint by continuity, and the whole trajectory lies in $\mathcal{K}_G$.

On the full-row-rank set, the formula $DH^\dagger = DH^\top(DH DH^\top)^{-1}$ and $H \in C^2$ show that b_ξ is C^1 . Since $\mathcal{K}_G$ is compactly contained in $\mathcal{U}_Q$, the standard ODE continuation and parameter-dependence theorem, applied along the compact trajectory, extends the ambient local flow and its C^1 dependence on initial data through $t = 1$. $\square$

Lemma 5.4 (Derivative of the full-row-rank right inverse). *For full-row-rank $B \in \mathbb{R}^{r \times d}$, let $A = B^\dagger = B^\top(BB^\top)^{-1}$. For perturbation E ,*

$$DA[B][E] = -AEA + (I_d - AB)E^\top A^\top A.$$

Under Assumption 5.2,

$$\|DA(v)[h]\|_{\text{op}} \leq C_A \|h\|, \quad C_A = \frac{2L_H}{s_0^2}.$$

Proof. Differentiate $B^\top(BB^\top)^{-1}$, use $D(C^{-1})[K] = -C^{-1}KC^{-1}$, and regroup using $BA = I_r$ and symmetry of AB . Because AB is an orthogonal projector and $\|A\| = 1/\sigma_r(B)$, each of the two terms is bounded by $\|E\|/\sigma_r(B)^2$. Substitute $E = D^2H(v)[h]$. $\square$

Theorem 5.5 (Quantitative fiber transport). *Under Assumption 5.2, for $z, z' \in G$ put $\xi = z' - z$ and solve*

$$\dot{v}(t) = DH(v(t))^\dagger \xi, \quad v(0) = v.$$

Each S_z is a compact neat C^2 manifold with boundary. The time-one map $\Phi_{z,z'} : S_z \rightarrow S_{z'}$ is a C^1 diffeomorphism with inverse $\Phi_{z',z}$. Writing $\Delta = \|z' - z\|$,

$$\|\Phi_{z,z'} - \text{Id}\|_\infty \leq \Delta/s_0, \tag{8}$$

$$\|D\Phi_{z,z'}\|_{\text{op}} \leq e^{C_A \Delta}, \tag{9}$$

$$\|D\Phi_{z,z'} - I_d\|_{\text{op}} \leq e^{C_A \Delta} - 1, \tag{10}$$

$$d_H(S_z, S_{z'}) \leq \Delta/s_0. \tag{11}$$

The differential maps $T_v S_z$ onto $T_{\Phi(v)} S_{z'}$, and

$$e^{-kC_A \Delta} \leq J_{\text{fib}} \Phi_{z,z'}(v) \leq e^{kC_A \Delta}. \tag{12}$$

Proof. Lemma 5.3 keeps every trajectory in $\mathcal{K}_G$, supplies its ambient C^1 local flow through time one, and justifies $DH(v)DH(v)^\dagger = I_r$ along the whole path. Consequently

$$\frac{d}{dt}H(v(t)) = \xi, \quad H(v(t)) = z + t\xi.$$

The boundary-submersion part of Assumption 5.2 gives the asserted neat level sets. Reversing ξ reverses the trajectory by uniqueness, proving the inverse formula. Since $\|DH^\dagger\| \leq 1/s_0$, integration proves (8), and applying it in both directions proves (11).

Lemma 5.4 bounds the vector-field derivative by $C_A \Delta$. The variational equation for $Y(t) = D\Phi_t(v)$ and Gronwall's inequality give $\|Y(t)\| \leq e^{C_A \Delta t}$, which proves (9). Its integral form also gives

$$\|Y(1) - I_d\| \leq \int_0^1 C_A \Delta e^{C_A \Delta t} dt = e^{C_A \Delta} - 1,$$

proving (10). Differentiating $H(\Phi(v)) = H(v) + \xi$ carries $\ker DH(v)$ into $\ker DH(\Phi(v))$; invertibility gives equality. Apply the derivative bound to both flow directions. Every tangential singular value then lies between $e^{-C_A\Delta}$ and $e^{C_A\Delta}$; their product is (12). $\square$

For $H(x, y) = ax$, fiber displacement is exactly $|z - z'|/|a|$, so the $1/s_0$ dependence cannot be removed.

6 Conditional measures and their transport

Define

$$J_H(v) = \sqrt{\det(DH(v)DH(v)^\top)}, \quad Q(z) = \int_{S_z} \frac{\rho(v)}{J_H(v)} d\mathcal{H}^k(v).$$

Under Assumption 5.2, the coarea theorem gives a canonical fiber law

$$\nu_z(dv) = \frac{\rho(v)J_H(v)^{-1}}{Q(z)} d\mathcal{H}^k|_{S_z}(v), \quad (13)$$

which agrees with $\mathcal{L}(V | H(V) = z)$ for $H_\# \mathcal{L}(V)$ -almost every z . This formula is standard and is not claimed as a contribution [1].

Theorem 6.1 (Conditional-measure transport). *Let $\tilde{\nu}_{z'} = (\Phi_{z,z'})_\# \nu_z$ and set*

$$L_w = \frac{L_\rho}{\rho_{\min}} + \frac{rL_H}{s_0}, \quad C_0 = \frac{L_w}{s_0} + kC_A.$$

Under Assumption 5.2, for $\Delta = \|z' - z\|$,

$$\left\| \log \frac{d\nu_{z'}}{d\tilde{\nu}_{z'}} \right\|_\infty \leq 2C_0\Delta, \quad (14)$$

$$\|\nu_{z'} - \tilde{\nu}_{z'}\|_{\text{TV}} \leq \tanh(C_0\Delta), \quad (15)$$

$$W_1(\nu_z, \nu_{z'}) \leq \frac{\Delta}{s_0} + D_K \tanh(C_0\Delta). \quad (16)$$

For $p \geq 1$,

$$W_p(\nu_z, \nu_{z'}) \leq \frac{\Delta}{s_0} + D_K \tanh(C_0\Delta)^{1/p}.$$

No Wasserstein-2 Lipschitz conclusion is asserted from this estimate.

Proof. Put $w = \rho/J_H$. Matrix differentiation gives

$$D \log J_H(v)[h] = \text{tr}(DH(v)^\dagger D^2 H(v)[h]).$$

The rank bound for the trace and the density lower bound imply $\|\nabla \log J_H\| \leq rL_H/s_0$ and $\|\nabla \log \rho\| \leq L_\rho/\rho_{\min}$. Along the horizontal trajectory $v(t)$ from v to $\Phi(v)$, rather than along an ambient line segment,

$$|\log w(\Phi(v)) - \log w(v)| \leq \int_0^1 \|\nabla \log w(v(t))\| \|\dot{v}(t)\| dt \leq \frac{L_w}{s_0} \Delta.$$

No global Lipschitz conclusion on the possibly nonconvex or disconnected set $\mathcal{K}_G$ is used. Together with (12), this yields

$$e^{-C_0\Delta} \leq \frac{w(\Phi(v))J_{\text{fib}}\Phi(v)}{w(v)} \leq e^{C_0\Delta}.$$

Fiber change of variables gives

$$Q(z') = \int_{S_z} w(\Phi(v)) J_{\text{fib}} \Phi(v) d\mathcal{H}^k(v),$$

so the same bounds hold for $Q(z')/Q(z)$. At $y = \Phi(v)$,

$$\frac{d\nu_{z'}}{d\tilde{\nu}_{z'}}(y) = \frac{w(y) J_{\text{fib}} \Phi(v) Q(z)}{w(v) Q(z')}.$$

This proves (14). Put $f = d\nu_{z'}/d\tilde{\nu}_{z'}$ and $a = 2C_0\Delta$. Then $e^{-a} \leq f \leq e^a$ and $\int f d\tilde{\nu}_{z'} = 1$. For every $x \in [e^{-a}, e^a]$,

$$\frac{|x - 1|}{x + 1} \leq \frac{e^a - 1}{e^a + 1} = \tanh(a/2).$$

Multiplying by $f + 1$, integrating, and using the half- L^1 convention for total variation proves (15).

Couple $v \sim \nu_z$ with $\Phi(v)$ to obtain $W_p(\nu_z, \tilde{\nu}_{z'}) \leq \Delta/s_0$. On a set of diameter $D_{\mathcal{K}}$, $W_p \leq D_{\mathcal{K}} \text{TV}^{1/p}$. The triangle inequality proves (16) and the general p bound. $\square$

The TV distance in (15) compares laws after putting them on the same transported fiber. Direct TV continuity between ν_z and $\nu_{z'}$ in ambient space is generally false because distinct fibers are disjoint.

Corollary 6.2 (Conditional data-law regularity). *For $\mu_z = F_{\#}\nu_z$,*

$$W_1(\mu_z, \mu_{z'}) \leq \frac{M_F}{s_0} \Delta + D_F \tanh(C_0\Delta).$$

A random-mask version follows by conditioning separately on each mask, provided that each conditional prior $V \mid M = m$ and map H_m satisfy Assumption 5.2, including its positive C^1 density hypothesis. The constants may depend on the mask.

Proof. Split through $F_{\#}\tilde{\nu}_{z'}$. The coupling $v \mapsto \Phi(v)$ gives

$$\begin{aligned} W_1(F_{\#}\nu_z, F_{\#}\tilde{\nu}_{z'}) &\leq \int_{S_z} \|F(\Phi(v)) - F(v)\| \nu_z(dv) \\ &\leq M_F \int_{S_z} \int_0^1 \|\dot{v}(t)\| dt \nu_z(dv) \leq \frac{M_F}{s_0} \Delta. \end{aligned}$$

Both $F_{\#}\tilde{\nu}_{z'}$ and $F_{\#}\nu_{z'}$ are supported in the set $F(\mathcal{K}_G)$ of diameter D_F . Hence their Wasserstein distance is at most D_F times their total-variation distance. Total variation contracts under pushforward, so (15) and the triangle inequality prove the claim. $\square$

Remark 6.3 (Why an ambient derivative bound is not a global Lipschitz bound). *On two disconnected components, a C^1 map may be constant with different values on the two components. Then $\sup \|DF\| = 0$ locally on the domain while the global Lipschitz constant is positive. This is why the transported-motion term above uses M_F only along actual flow trajectories, whereas the reweighted-mass term uses the image diameter D_F .*

7 Image manifolds and Gaussian smoothing

Assume $k = d - r \geq 1$ and suppose the intrinsic tangent noncollapse condition

$$\tau_0 = \inf_{v \in \mathcal{K}_G} \inf_{\substack{u \in \ker DH(v) \\ \|u\|=1}} \|DF(v)u\| > 0 \quad (17)$$

and $F|_{S_z}$ is injective for every $z \in G$.

Theorem 7.1 (Motion and curvature of conditional image manifolds). *Under Assumption 5.2, (17), and the stated injectivity of $F|_{S_z}$ for every $z \in G$, $\mathcal{M}_z = F(S_z)$ is an embedded C^2 manifold, possibly with boundary, of dimension k . The conjugated map*

$$\Psi_{z,z'} = F \circ \Phi_{z,z'} \circ (F|_{S_z})^{-1}$$

is a C^1 diffeomorphism $\mathcal{M}_z \rightarrow \mathcal{M}_{z'}$ and

$$\|\Psi_{z,z'} - \text{Id}\|_\infty \leq (M_F/s_0)\Delta, \quad (18)$$

$$\|D\Psi_{z,z'} - I\|_{T\mathcal{M}_z \rightarrow \mathbb{R}^D} \leq \frac{1}{\tau_0} \left[M_F(e^{C_A\Delta} - 1) + \frac{M_{F,2}\Delta}{s_0} e^{C_A\Delta} \right]. \quad (19)$$

Moreover,

$$\|\Pi_{\mathcal{M}_z}\|_{\text{op}} \leq \frac{M_{F,2} + M_F L_H / s_0}{\tau_0^2}.$$

Proof. Tangent noncollapse makes $F|_{S_z}$ an immersion. A continuous injective immersion from compact S_z to Euclidean space is an embedding. Conjugating the fiber diffeomorphism proves the first claim and the displacement bound. For $u \in T_v S_z$,

$$D\Psi(DF(v)u) - DF(v)u = [DF(\Phi(v))D\Phi(v) - DF(v)]u.$$

Along the horizontal trajectory from v to $\Phi(v)$,

$$DF(\Phi(v)) - DF(v) = \int_0^1 D^2 F(v(t))[\dot{v}(t)] dt, \quad \|DF(\Phi(v)) - DF(v)\| \leq \frac{M_{F,2}}{s_0} \Delta.$$

Combining this bound with (9)–(10) and $\|(DF|_{T_v S_z})^{-1}\| \leq 1/\tau_0$ gives the derivative estimate without requiring the Euclidean chord from v to $\Phi(v)$ to remain in the regular region.

Differentiating H twice in a fiber chart gives

$$\Pi_{S_z}(u, w) = -DH(v)^\dagger D^2 H(v)[u, w],$$

hence $\|\Pi_{S_z}\| \leq L_H/s_0$. The second derivative of $F|_{S_z}$ is bounded by $M_{F,2} + M_F L_H/s_0$. Unit image tangents lift with norm at most $1/\tau_0$, and normal projection does not increase norm, proving the curvature bound. $\square$

Curvature does not imply a positive uniform reach: distant parameter points can have nearby images. No reach claim is made without global separation.

Proposition 7.2 (Gaussian-smoothed conditional density). *For $\alpha \in \mathbb{R}$, $\sigma > 0$, and the centered Gaussian density*

$$\varphi_\sigma(x) = (2\pi\sigma^2)^{-D/2} \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right),$$

define

$$p_\sigma(x | z) = \int \varphi_\sigma(x - \alpha y) \mu_z(dy).$$

Then

$$\|p_\sigma(\cdot | z) - p_\sigma(\cdot | z')\|_1 \leq \sqrt{2/\pi} \frac{|\alpha|}{\sigma} W_1(\mu_z, \mu_{z'}), \quad (20)$$

$$\|\nabla p_\sigma(\cdot | z) - \nabla p_\sigma(\cdot | z')\|_1 \leq \frac{|\alpha| \sqrt{D+1}}{\sigma^2} W_1(\mu_z, \mu_{z'}). \quad (21)$$

Proof. For any coupling (Y, Y') , translate the Gaussian along the segment between αY and $\alpha Y'$. Fubini's theorem and rotational invariance give

$$\|\varphi_\sigma(\cdot - \alpha Y) - \varphi_\sigma(\cdot - \alpha Y')\|_1 \leq |\alpha| \|Y - Y'\| \sqrt{2/\pi} / \sigma.$$

For gradients, use the same argument with $D^2\varphi_\sigma(x)u$. If $G \sim \mathcal{N}(0, I_D)$, Cauchy-Schwarz yields

$$\int \|D^2\varphi_\sigma(x)u\| dx \leq \sigma^{-2} \left[\mathbb{E} \|(GG^\top - I)u\|^2 \right]^{1/2} = \sqrt{D+1}/\sigma^2.$$

Average and minimize over couplings. □

These estimates do not prove score regularity for $\nabla p_\sigma/p_\sigma$; Gaussian tails create a denominator problem.

8 Conditional diffusion entropy corollary

Corollary 8.1 (Version-pinned pointwise DDPM entropy input). *Use the random-mask setting and the Borel set Z_{reg} of Corollary 3.4. For almost every fixed $(m, z) \in Z_{\text{reg}}$ for which a positive certified covering exponent $k_{\text{ent}}(m, z) \geq 1$ is selected, the default $2n_v - r(m, z)$ is available when positive, while $\ell(m, z)$ is available under the additional constant image-rank hypothesis in that corollary. Here k_{ent} is a parameter in the external single-scale covering assumption, not a claim that this upper exponent is the minimal Minkowski dimension. Assume the remaining conventions and hypotheses of Theorem 2 of Liang et al. [11, arXiv version 2], and let (X_t) and (Y_t) denote, respectively, the forward and original-DDPM chains defined there. In the notation of that theorem, assume the conditional score errors*

$$\varepsilon_{\text{score},t}^2(m, z) = \mathbb{E}[\|s_t(X_t; m, z) - s_t^*(X_t; m, z)\|^2 | Z = (m, z)]$$

satisfy its unweighted condition

$$\frac{1}{T} \sum_{t=1}^T \varepsilon_{\text{score},t}^2(m, z) \leq \varepsilon_{\text{score}}^2(m, z).$$

Then, for all sufficiently large T (with the threshold allowed to depend on the fixed observation),

$$\text{TV}(p_{X_1|Z=(m,z)}, p_{Y_1|Z=(m,z)}) \lesssim_{m,z} \frac{k_{\text{ent}}(m, z) \log^3 T}{T} + \varepsilon_{\text{score}}(m, z) \sqrt{\log T}.$$

Proof. Fix such a pair (m, z) outside the exceptional null set, abbreviate its selected positive exponent by $k \geq 1$, and choose the single scale $\varepsilon_0 = T^{-c_{\varepsilon_0}}$ required by Assumption 1 of Liang–Huang–Chen v2. Corollary 3.4 gives

$$\log \mathcal{N}_{\varepsilon_0} \leq C''_{m,z} + kc_{\varepsilon_0} \log T \leq \tilde{C}_{m,z} k \log T$$

for $T \geq 2$, after absorbing the fixed additive constant. This is their single-scale intrinsic-dimension condition. The conditional support is compact, so its fixed radius is at most T^{c_R} for all sufficiently large T , as required by their Assumption 2. The displayed score condition is their Assumption 3. Applying their Theorem 2 to this fixed conditional target law gives the stated pointwise bound. $\square$

The source EIT paper invokes the same single-scale result in its Theorem 4.15. We pin arXiv version 2 because its hypotheses are the ones verified in the preceding argument; adapting the corollary to subsequent formulations is left for future work. The result remains external and pointwise conditional: neither its score-error hypothesis nor constants uniform in (m, z) are established here. When either $r(m, z) = 2n_v$ or a certified restricted image rank $\ell(m, z) = 0$, the conditional support is finite, but we do not substitute a zero-dimensional parameter into an external theorem formulated for positive intrinsic dimension.

9 Discretized EIT model and numerical methodology

The numerical examples use a unit-disk finite-element model with piecewise linear basis functions. A conductivity equals one outside a convex quadrilateral and four inside it. The four counterclockwise vertices are indexed by a fixed template convention, giving a parameter in $\mathbb{R}^8$. The admissible generator enforces positive margins for side length, distance from the disk boundary, and interior angles. This finite-dimensional chart is the object differentiated numerically; the experiments do not establish the regularity of the corresponding continuum shape map [2].

The principal mesh has 145 vertices, 256 triangles, and 32 boundary nodes. The discrete DtN matrix is symmetric, and its 528 upper-triangular entries form the complete data vector. Entrywise background normalization fixes the observation metric. A mask selects rows of the resulting Jacobian; all reported singular values therefore refer to the Euclidean vertex metric and the declared normalized or noise-whitened observation metric. The synthetic population was generated with seed 20260901 and separated into 2,000 training, 200 validation, and 400 test polygons. The full population data are not redistributed in this repository; the retained numerical files are compact, non-personal diagnostic outputs.

To examine the image-rank refinement without inferring an exact rank, a post-hoc audit recomputed the full singular spectrum of 360 archived 528×8 finite-difference Jacobians from three independent 120-polygon population studies (seeds 20260828–20260830). The same relative threshold is used as below. This pointwise diagnostic does not check constant rank on a neighborhood of an observation fiber.

Central differences use the nondimensionalized parameter step $h = 0.002$ and a relative singular-value threshold of 10^{-3} unless stated otherwise. A step-size sweep from 10^{-2} to 10^{-6} checks the stability of this choice. Candidate masks are selected without validation or test polygons. The fiber experiments use two constructions: predictor–corrector continuation initialized in a numerical nullspace, and constrained correction from isotropic perturbations in the full $\mathbb{R}^8$ parameter space. The latter serves as an independent control on the dimension supplied to the first method. Local PCA uses a 10^{-2} relative eigenvalue threshold. Greedy covering estimates are calibrated on Euclidean balls of known dimension at identical sample sizes.

The learned-completion endpoint reuses a completed frozen protocol. One six-entry mask was selected from regularized Gram log-determinant scores computed on 80 training Jacobians, before network fitting. Specifically, the score averages $\log \det(J_M J_M^\top + 10^{-8} a_M I)$, where $a_M = \text{tr}(J_M J_M^\top)/6$, across the training Jacobians. The selected mask maximized this score among 2,000 random six-entry candidates; 20 controls were sampled from the middle 80% of the score ordering after excluding both extreme deciles. Three conditional EDMs and three deterministic U-Nets, with network seeds 20260911–20260913, were selected only through validation checkpoints. Evaluation used 400 held-out polygons, normalized noise standard deviations 0, 10^{-4} , 3×10^{-4} , and 10^{-3} , 18 EDM sampling steps, and 16 learned conditional samples. The EDM checkpoints at step 50,000 minimized validation EMA loss over 10,000-step candidates, and the U-Net checkpoints at step 10,000 minimized validation missing-entry MSE over 2,000-step candidates, for every seed. The evaluation seed was 20260921. The reported 95% intervals use 10,000 crossed bootstrap replicates (seed 20261001), resampling the network-seed and polygon axes while holding the prespecified masks and the already aggregated sample axis fixed.

The mesh comparison maps the same physical boundary-coordinate pairs to three refinements. A separate complete-electrode-model (CEM) derivative diagnostic uses reference software and mesh material distributed with the KTC 2023 v3 dataset [14]; only aggregate derivative diagnostics are retained here. No KTC code, mesh, or measurement data are redistributed. The source dataset and its license are identified in the repository’s third-party notice.

10 Numerical results

The experiments isolate predictions of the preceding analysis rather than testing a conductivity-reconstruction endpoint. They provide finite- dimensional numerical evidence and do not replace any regularity assumption or proof.

10.1 Independent fiber recovery and rank along sampled fibers

The first continuation method used predictors in the numerical nullspace of a base Jacobian. To remove construction-induced circularity, a second method started isotropically in the full $\mathbb{R}^8$ parameter space and corrected the observation residual without nullspace initialization. Both methods recovered the dimensions shown in Table 1.

Table 1: Independent local fiber recovery. PCA uses a 10^{-2} relative eigenvalue threshold.

thresholded rank	predicted $8 - r$	nullspace start	full- $\mathbb{R}^8$ start
2	6	6	6
4	4	4	4
6	2	2	2

The full-space experiment was repeated on ten held-out polygons stratified by admissibility margin. All 30 recovered clouds matched the predicted PCA dimension. The replicated clouds accepted 720 of 824 full-space starts. All 104 failures came from one near-boundary base polygon, whose recorded boundary margin was 3.23×10^{-4} . Including the paired generator controls and the rank-eight endpoint, 1,104 of 1,208 starts were accepted; adding the 34 base points gives the 1,138 points at which centered-difference Jacobians were recomputed. All 1,138 accepted/base points retained the expected thresholded full row rank. The earlier direct-continuation study separately

accepted 960 of 966 attempts; its six failures reached an iteration or line-search limit. Rejected candidate points, their singular values, and their admissibility margins were not archived. Thus the rank-stability statement is explicitly conditioned on acceptance and does not resolve selection bias. For the endpoint $r = d = 8$, 96 nearby full-space starts returned to the same numerical solution, with maximum distance 1.1609×10^{-7} from the base point, which is consistent with local isolation but does not establish it.

All 360 complete-data Jacobians in the post-hoc image-rank audit had numerical rank eight at the 10^{-3} relative threshold; the minimum and median values of σ_8/σ_1 were 0.0237 and 0.0865. Consequently, on these sampled points the numerical rank-difference formula gives $\ell = 8 - r$, not a stricter exponent. This negative diagnostic neither certifies exact full rank nor excludes another component or distant part of a fiber with different rank.

10.2 Full spectrum, criticality, covering bias, and mesh sensitivity

Across six right-singular directions, four residual radii, and both signs, the 48 quantities $\sigma_i W_i/\delta$ had mean 0.999961 and range $[0.998971, 1.000797]$. On 48 separate rank-six linear ellipsoids, the full-spectrum logarithmic cover predictor had mean usable Spearman correlation 0.990 with greedy cover count, versus 0.854 for a surrogate that repeats only $\sigma_{\min}$. At the eight increasing cover scales, respectively, 21, 13, 6, 4, 2, 1, 0, and 0 of the 48 empirical covers saturated at the 360-point cloud size. The declared criterion $0.1n < \text{median } N_\varepsilon < 0.9n$ retained the four coarsest scales; their per-scale correlations were 0.984–0.994 for the full spectrum and 0.828–0.881 for the surrogate. Because both predictors and the sampled ellipsoids use the same linear singular spectrum, this is a finite-cloud formula calibration, not an independent validation of nonlinear EIT tube geometry.

A same-rank near-critical mask had $\sigma_6 = 3.63 \times 10^{-4}$, versus 0.263 for the stable mask, and amplified mean linearized parameter error by a factor of 402.7. This is a near-critical stability control, not evidence of exact rank change or fiber branching.

Finite-cloud covering slopes are downward biased. Applying the identical estimator to 193 points from known Euclidean balls embedded in $\mathbb{R}^8$ gives Table 2; therefore raw EIT slopes are not treated as consistent continuum-dimension estimates.

Table 2: Greedy-cover calibration at 193 points; mean $\pm$ standard deviation over four independent clouds.

true dimension	estimated slope	downward bias
2	1.507 ± 0.037	24.6%
4	2.629 ± 0.093	34.3%
6	3.647 ± 0.242	39.2%

Mapping the same six physical boundary-coordinate pairs to three disk-mesh refinements preserved thresholded rank six. The normalized weakest-direction widths were 0.9986, 0.9996, and 0.9988; the normalized spectrum changed by 0.0819 between the primary and finest mesh. This is a discretization sensitivity check, not a continuum FEM convergence theorem.

10.3 Learned conditional completion

Table 3 reports the independently regenerated summary of the frozen learned study. At zero added noise, the selected-mask EDM missing-entry error was 0.0271, compared with 0.0483 averaged across the 20 score-trimmed random controls. The control/selected ratio was 1.781 with crossed

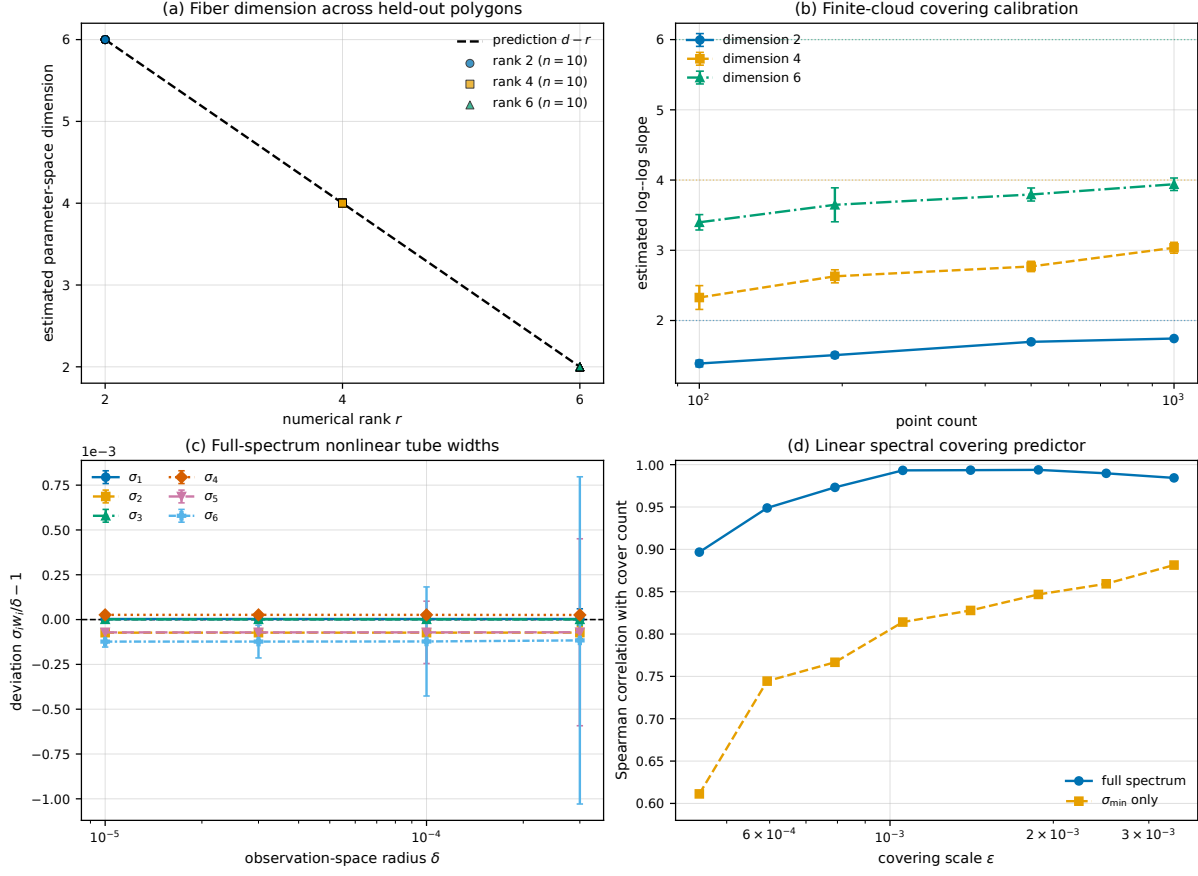


Figure 1: Numerical tests of the local geometric mechanisms. Top left: independent full-dimensional fiber recovery for ten held-out polygons; top right: finite-cloud covering calibration; bottom left: deviations of the normalized singular-direction widths $\sigma_i W_i / \delta$ from one; bottom right: full-spectrum and weakest-singular-value covering predictors.

seed-polygon 95% interval $[1.709, 1.856]$. The corresponding projected energy-score ratio was 1.826 with interval $[1.754, 1.904]$.

At zero noise, the U-Net selected and random errors were 0.0301 and 0.0402 (ratio 1.334). Projected 90% interval coverage was 0.862 for the selected mask and 0.838 across control masks. Among the 20 control masks, the Spearman correlation between the training-only regularized Gram log-determinant score and held-out EDM error was -0.812 , with crossed-bootstrap interval $[-0.847, -0.761]$. The physical row-sum projection reduced the selected-mask residual by a factor of 8,594, while its projected/unprojected error ratio was 0.99982. The large residual factor is expected because the affine projection enforces the row-sum constraint by construction and leaves a denominator near numerical precision; it is not independent evidence of learned accuracy.

Table 3: Held-out conditional EDM results. Error ratios are control/selected; their intervals use crossed network-seed–polygon bootstrap resampling.

noise s.d.	EDM selected	EDM control	error ratio [95% CI]	projected energy-score ratio
0	0.0271	0.0483	1.781 [1.709, 1.856]	1.826
10^{-4}	0.0274	0.0483	1.762 [1.688, 1.842]	1.792
3×10^{-4}	0.0273	0.0484	1.777 [1.706, 1.854]	1.822
10^{-3}	0.0271	0.0484	1.781 [1.707, 1.864]	1.821

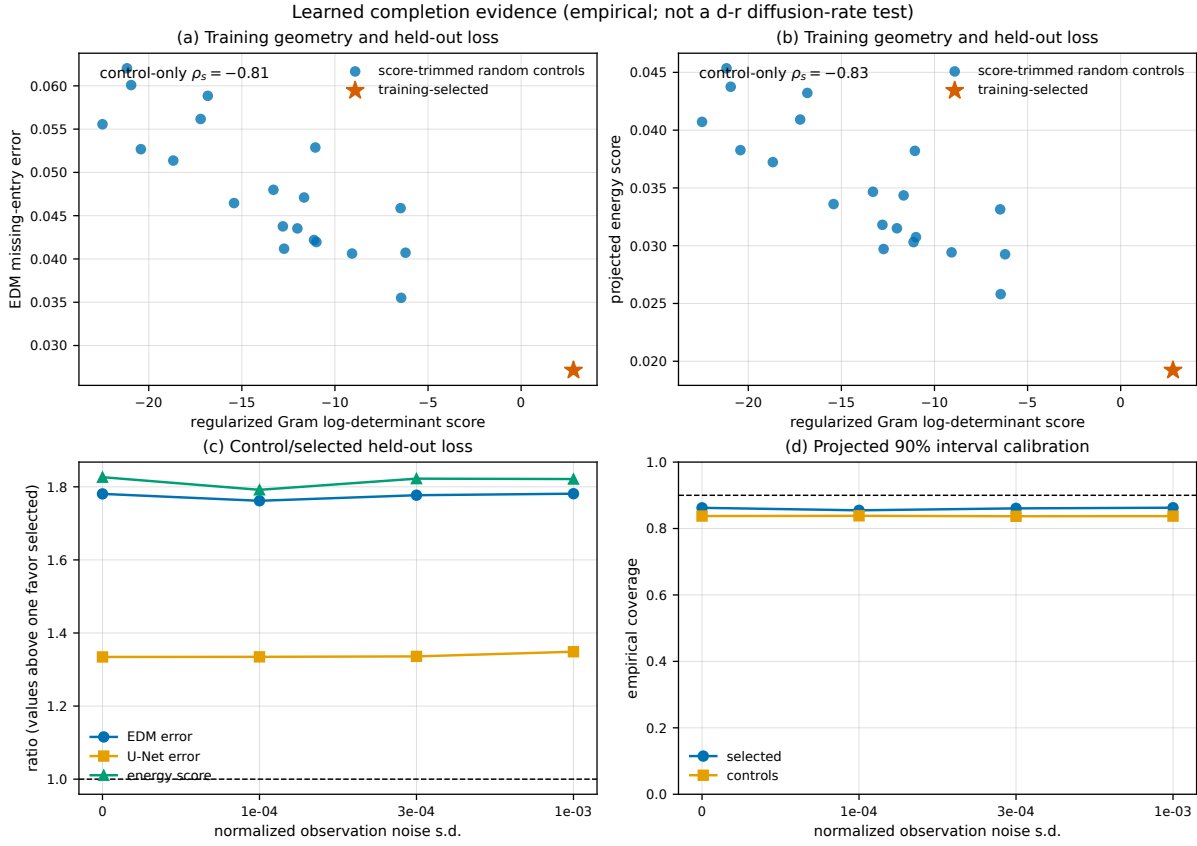


Figure 2: Held-out learned-completion evidence. Panels (a)–(b) relate the training-only spectral mask score to test losses across control masks, with the selected mask shown separately. Panel (c) compares control/selected losses across noise levels, and panel (d) reports interval coverage. This experiment uses equal-budget masks and EDM sampling; it is not a test of the pointwise DDPM entropy corollary or of an empirical $d - r$ learning rate.

All masks observe six entries. The selected mask has mean training numerical rank six, while the control-mask means range from 5.65 to 6.00. Thus this study supplies an actual conditional-generation endpoint and links a training-only full-spectrum score to held-out performance, but it does not separate residual dimension from conditioning, establish score approximation, or show that the entropy refinement reduces network sample complexity, training cost, or diffusion steps.

11 Supporting identities and interface checks

The following results are standard supporting tools. They are included for completeness and are not presented as contributions.

Proposition 11.1 (Nonexpansion of compatible affine projection). *Let $\mathcal{C}$ be a nonempty closed affine subset of a finite-dimensional Hilbert space and let $\Pi_{\mathcal{C}}$ be its orthogonal projection. If the truth $X^* \in \mathcal{C}$, then every estimate $\hat{X}$ satisfies*

$$\|\Pi_{\mathcal{C}}\hat{X} - X^*\| \leq \|\hat{X} - X^*\|.$$

Proof. Orthogonal projection onto a nonempty closed convex set obeys the variational inequality $\langle \hat{X} - \Pi_{\mathcal{C}}\hat{X}, X^* - \Pi_{\mathcal{C}}\hat{X} \rangle \leq 0$. Expanding the squared distance from $\hat{X}$ to X^* gives the result. $\square$

In 20,000 compatible random cases, the numerical inequality held to the stated tolerance. In a noisy incompatible control, hard projection either failed to improve the distance or violated compatibility in all 20,000 cases. The control illustrates why compatibility of the truth is required.

Proposition 11.2 (Linear-Gaussian missing-response MMSE). *Let $\eta \sim \mathcal{N}(0, I_{\ell})$, $y = \mu + B\eta$, and observe*

$$\tilde{y}_M = \mu_M + B_M\eta + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \Gamma_M),$$

independently, with Γ_M positive definite. Put

$$P_M = (I_{\ell} + B_M^{\top}\Gamma_M^{-1}B_M)^{-1}.$$

Then the Bayes risk for squared-error prediction of the missing response is

$$R_M = \mathbb{E}\|y_{\bar{M}} - \mathbb{E}[y_{\bar{M}} \mid \tilde{y}_M]\|^2 = \text{tr}(B_{\bar{M}}P_MB_{\bar{M}}^{\top}).$$

Proof. Gaussian conditioning, or completion of the square, gives posterior covariance $\text{Cov}(\eta \mid \tilde{y}_M) = P_M$. The posterior covariance of $y_{\bar{M}}$ is therefore $B_{\bar{M}}P_MB_{\bar{M}}^{\top}$. The conditional mean is the squared-error Bayes estimator, and its mean squared error is the trace of this covariance. $\square$

With 200,000 samples, the theoretical risk was 0.1307554 and the empirical risk was 0.1312530, a relative difference of 0.381% and a Monte Carlo z -score of 1.39.

Proposition 11.3 (Smooth finite-dimensional CEM voltage chart). *Let a bounded Lipschitz domain have fixed disjoint electrodes of positive surface measure and positive contact impedances. Suppose $\theta \mapsto \gamma_{\theta}$ is C^1 from an open subset of $\mathbb{R}^p$ to L^{∞} , with conductivity uniformly bounded above and away from zero on compact parameter subsets. For fixed gauge-normalized current patterns, the finite vector of complete-electrode-model voltages is C^1 in θ .*

Proof. On the gauge-fixed space $\mathcal{H} = H^1(\Omega) \oplus \mathbb{R}_0^L$, the standard CEM bilinear form is

$$a_\theta((u, U), (v, V)) = \int_{\Omega} \gamma_\theta \nabla u \cdot \nabla v \, dx + \sum_{j=1}^L z_j^{-1} \int_{E_j} (u - U_j)(v - V_j) \, ds.$$

The conductivity and impedance bounds, trace theorem, and gauge condition give uniform boundedness and coercivity on compact parameter subsets [15]. Its operator $A(\theta) : \mathcal{H} \rightarrow \mathcal{H}^*$ is C^1 . The identity $D(A^{-1})[h] = -A^{-1}(DA[h])A^{-1}$ makes the weak solution C^1 , and finite voltage readout is bounded linear. $\square$

The implemented four-dimensional CEM chart had a finite-difference plateau change below 1.16×10^{-5} , relative discrepancy 0.00124 from the reference-solver Jacobian at the reference step, and directional Taylor residual slope 1.976. This checks the software interface in a smooth coefficient chart; it does not establish shape differentiability for moving discontinuous polygons.

The anisotropic whitening control used 500 masks and 400 trials per mask. The Spearman correlation between inverse singular value and observed amplification was 0.413 in raw coordinates and 0.988 after whitening. In a separate selection study, the unweighted design error was 3.93 times the whitened design error. These experiments support the whitening prescription in Corollary 4.2; they do not make whitening a new theorem.

12 Analytic sanity checks

The numerical package tests four transparent cases. For $H(x, y) = x$, the transport is $(z, y) \mapsto (z', y)$ and the motion bound is exact with $s_0 = 1$. For $H(x, y) = ax$, displacement is $|z - z'|/|a|$. For $H(x, y) = x + y^2$,

$$DH(x, y)^\dagger = \frac{(1, 2y)^\top}{1 + 4y^2},$$

and the horizontal ODE preserves $H(v(t)) - t\xi$ exactly. Finally, $H(x, y) = x^2$ has $s_0 = 2|x|$; the lift norm diverges at $x = 0$, the rank collapses, and no uniform Lipschitz transport crosses the critical value.

13 Discussion and limitations

The mathematical results form the following chain. Exact conditioning localizes the population law to a compact subset of a $(d-r)$ -dimensional regular level set; under the stated constant-rank hypotheses, restricted image rank can sharpen its image covering exponent to $\ell = \text{rank } DF - \text{rank } DH$; noisy finite-resolution complexity resolves the complete whitened singular spectrum; quantitative horizontal flow transports regular fibers; and coarea weighting turns this into conditional-law Wasserstein-1 regularity. At the single scale required by the explicitly pinned arXiv-v2 external DDPM theorem, the entropy refinement supplies its covering input for almost every fixed regular observation for which a positive certified exponent is selected, conditional on that theorem's other hypotheses. The EIT experiments are consistent with the predicted local mechanisms, and the separate learned study supplies a conditional EDM completion endpoint without testing that DDPM theorem.

The result is deliberately narrower than a complete conditional-diffusion theory. The project has not established whole-prior EIT flow completeness, uniform reach, conditional-score regularity, or finite-sample score estimation. Those items are excluded from the claim set and recorded only in the repository future-work document.

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